

# Matrices and Determinants

Rajat Kalia

19 August, 2026

## Infrastructure

#764, 41-A, Chandigarh {TC(official)} - 160036  
 Shamlat (Ramgarh Sardaran), Ludhiana, Punjab  
 SCO 24(TF), 41-D, Chandigarh - 160036  
<https://www.alphaclasses.com/>

## Educational

<https://github.com/rajatkalia1982>  
<https://sourceforge.net/u/rajatkalia/profile>  
<https://www.youtube.com/@rajatkaliaeducation>

## Store

<https://www.lulu.com/spotlight/rajatkalia11>  
<https://amazon.com/author/rajatkalia>

## Sports

<https://www.youtube.com/@indiangamersorganization>  
<https://www.facebook.com/groups/igohn>  
<https://india.voobly.com/>

## Social

<https://www.goodreads.com/sardarbhagatsingh>  
<https://x.com/RajatKalia1>  
<https://www.linkedin.com/in/rajatkalia007/>  
<https://www.facebook.com/rajatkalia007>  
<https://www.instagram.com/rajatkalia007/>  
<https://www.twitch.tv/rajatkalia>  
<https://www.facebook.com/rajatkaliaeducation>  
<https://www.researchgate.net/profile/Rajat-Kalia>  
<https://www.w3profile.com/rajatkalia/>  
<https://www.reddit.com/user/rajatkalia/>

## Blogs

<https://rk-e.blogspot.in/> [Physics]  
<https://chem-rke.blogspot.com/> [Chemistry]  
<https://biosciences-rke.blogspot.com/> [ Bio-Sciences ]  
<https://rk-edu.blogspot.com/> [ Mathematics ]  
<https://medlife-rke.blogspot.com/> [Medical Sciences ]  
<https://computers-mke.blogspot.com/> [ Computers Engineering ]  
<https://ee-mke.blogspot.com/> [Electrical Engineering ]  
<https://me-mke.blogspot.com/> [ Mechanical Engineering ]  
<https://ce-mke.blogspot.com/> [ Civil Engineering ]  
<https://mke-whj.blogspot.com/> [ White Hat Junior ]  
<https://ba-xpn.blogspot.com/> [ Bachelor of Arts ]  
<https://constitution1982.blogspot.com/> [ Constitution ]  
<https://debk-mke.blogspot.com/> [ Accounting ]  
<https://comm-pke.blogspot.com/> [ Commerce ]  
<https://brigade-13.blogspot.com/> [ Air Force, India ]  
<https://battalion-7.blogspot.com/> [ Armed Forces ]  
<https://design-mke.blogspot.com/> [ Design ]  
<https://tmm18.blogspot.com/> [ Meditation ]

## Freelancing

<https://www.guru.com/freelancers/rajat-kalia>  
<https://www.codecademy.com/profiles/rajatkalia>  
<https://leetcode.com/u/rajatkalia1/>  
<https://www.scribd.com/user/308680256/Rajat-Kalia>

# Contents

|          |                                               |          |
|----------|-----------------------------------------------|----------|
| <b>I</b> | <b>Matrices Problem Set</b>                   | <b>3</b> |
| <b>1</b> | <b>Matrices</b>                               | <b>4</b> |
| 1.0.0.1  | Linked Comprehension Type Questions . . . . . | 5        |
| 1.0.0.2  | Matrix Match Type Problems . . . . .          | 5        |
| <b>2</b> | <b>JEE Mains Problems</b>                     | <b>6</b> |

Part I

Matrices Problem Set

# Chapter 1

## Matrices

**Q:** Invent a  $3 \times 3$  magic matrix M with the entries 1 , 2 , 3 , .....9. All rows and columns and diagonals add to 15 . The first row could be ( 8, 3, 4) . What could be M.

**Q:** If  $A = \begin{bmatrix} 1 & i \\ -i & 1 \end{bmatrix}$  . Find  $A^T A$ .

**Q:** Find the adjoints of the following matrices

a)  $\begin{bmatrix} 15 & 7 \\ 8 & 14 \end{bmatrix}$

b)  $\begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}$

c)  $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

**Q:** If  $A = \begin{bmatrix} 2x+6 \\ 26 \\ 2z+48 \\ t^2+41 \end{bmatrix}$

$$= \begin{bmatrix} x^2 & -x & 1 \\ y^2 & -y & 2 \\ z^2 & -z & 3 \\ t^2 & -t & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 3 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Then , A can be

a)  $\begin{bmatrix} 10 \\ 26 \\ 46 \\ 47 \end{bmatrix}$

b)  $\begin{bmatrix} 10 \\ 26 \\ 54 \\ 42 \end{bmatrix}$

c)  $\begin{bmatrix} 10 \\ 26 \\ 46 \\ 42 \end{bmatrix}$

d)  $\begin{bmatrix} 10 \\ 26 \\ 54 \\ 41\frac{9}{25} \end{bmatrix}$

### 1.0.0.1 Linked Comprehension Type Questions

**Comprehension 1:** Two vectors  $v = (1, 2, 3)$  and  $w = (1, 3, 4)$  are taken and put in the columns of a matrix as

$$M = \begin{bmatrix} 1 & 1 \\ 2 & 3 \\ 3 & 4 \end{bmatrix}$$

The linear combination of these two vectors can be expressed as

$$c \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + d \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 2 & 3 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} A \\ B \end{bmatrix}$$

Q1: A relation between A , B , c and d is

- a)  $6c + 8d = A + B$
- b)  $8c + 6d = A + B$
- c)  $6c + 8d = 6A + 8B$
- d) None of these

Q2: The matrix  $\begin{bmatrix} A \\ B \end{bmatrix}$  in the equation

$$\begin{bmatrix} 1 & 1 \\ 2 & 3 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} A \\ B \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 8 \end{bmatrix} \text{ is given by}$$

- a)  $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$
- b)  $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$
- c)  $\begin{bmatrix} 2 \\ 2 \end{bmatrix}$
- d) None of these

### 1.0.0.2 Matrix Match Type Problems

**Matrix 1 :** Three elementary matrices  $E = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ ,  $F = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$ ,  $G = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$  are given. Under column I , resultant matrices (lower triangular) obtained from  $E, F, G$  or their inverses are given. Match the matrices given in column I with their corresponding expressions given under column II :

| Column I                                                                  | Column II                |
|---------------------------------------------------------------------------|--------------------------|
| (P) $\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$ | (A) $EFG$                |
| (Q) $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix}$ | (B) $EF^{-1}G$           |
| (R) $\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$  | (C) $E^{-1}FG^{-1}$      |
| (S) $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & -1 & 1 \end{bmatrix}$  | (D) $E^{-1}F^{-1}G^{-1}$ |
|                                                                           | (E) $G^{-1}F^{-1}E^{-1}$ |

## Chapter 2

### JEE Mains Problems

Q1 : Let  $A = \begin{bmatrix} 1 & 2 \\ 1 & \alpha \end{bmatrix}$  and  $B = \begin{bmatrix} 3 & 3 \\ \beta & 2 \end{bmatrix}$ . If  $A^2 - 4A + I = 0$  and  $B^2 - 5B - 6I = 0$ , then among the two statements :

$$(S1) : [(B - A)(B + A)]^T = \begin{bmatrix} 13 & 15 \\ 7 & 10 \end{bmatrix}$$

$$(S2) : \det(\operatorname{adj}(A + B)) = -5,$$

[ JEE Mains 2nd Apr 2026 Shift 1 Q4 ]

(A) Only (S1) is correct

(B) Only (S2) is correct

(C) Both (S1) and (S2) are correct

(D) Both (S1) and (S2) are wrong

{Solution : <https://www.youtube.com/watch?v=uTUbsovqU8>}

# Index